

Love AI More. Trust AI Less.

DSCT lecture walkthrough

1 Claim, evidence, and reading

Claim: Increasing simultaneous requests should increase throughput until parallelism is saturated. The generated table and plot show a knee near the configured limit; serial rows mostly hide setup overhead.

2 Assumptions and caveats

- Token counts and per-request rates are roughly comparable.
- The filename-derived parallel limit is a control, not a directly instrumented server measurement.
- GPU placement is an outcome: the 3 GB card's limit-4 run was 84% GPU / 16% CPU.
- More hosts, repeated runs, and direct timing of generation would strengthen the conclusion.

3 The check

Compute overlap = aggregate rate/per-request rate. For the T4 at limit 8 and $N = 16$, $125.3/25.5 \approx 4.91$: about five requests were generating at once. For the same card at limit 1 and $N = 16$, $76.4/138.3 \approx 0.55$: no batching claim is justified.